Every transitive set sits inside a stage of the cumulative hierarchy

The Open Logic Project contributors

openlogic:set-theory/spine:local-snapshot-20261013; CC BY 4.0

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-set-theory-spine/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: both definitions, the D-empty reduction and the minimal-counterexample contradiction are verbatim in the span; the author's one-character display typo (lowercase b where B is meant) is preserved rather than silently corrected.

Editorial note. The label lem:TransitiveWellFounded names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lem environment in the unit source.

Allowed standard prerequisites (editorial).

- Strict supremum of a set of ordinals. Unit ordinals, definition defsupstrict (supremum not attained), cited with its marker at proof line 482. Shapes alpha as supstrict of the witnessing ordinals so that membership in alpha is what places elements into V_alpha.
- Basic properties of the V-hierarchy. Same unit, \ollabel{Valphabasicprops} (source line 847). Line 490: x subseteq V_delta with delta in alpha yields x in V_alpha via the cumulation clause.
- Foundation / set induction. ZFC axiom of Foundation as used at line 494: a non-empty D contains a member disjoint from D. The reductio paragraph: minimal counterexample B in D with D cap B empty.

Source reference. Original definition defsupstrict: sources/olp-set-theory-ordinals/pinned-original-source.tex, source lines 970–970. Named cross-unit citation: the target label is staged in the corpus under unit olp-set-theory-ordinals at source line 970; the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference Valphabasicprops: sources/olp-set-theory-spine/pinned-original-source.tex, source lines 847–847. Prerequisite pointer invoked at proof line(s) 490; it names a standard fact of the same unit that the

counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *If A is a transitive set, then there is some α such that $A \subseteq V_\alpha$.*

Proof. Recalling the definition of "$\subsetneq (X)$" from defsupstrict, define two sets:

$$D = \{\, x \in A \mid \forall \delta\ x \nsubseteq V_\delta \,\}$$
$$\alpha = \subsetneq \{\, \delta \mid (\exists x \in A)(x \subseteq V_\delta \wedge (\forall \gamma \in \delta) x \nsubseteq V_\gamma) \,\}$$

Suppose $D = \emptyset$. So if $x \in A$, then there is some δ such that $x \subseteq V_\delta$ and, by the well-ordering of the ordinals, $(\forall \gamma \in \delta) x \nsubseteq V_\gamma$; hence $\delta \in \alpha$ and so $x \in V_\alpha$ by Valphabasicprops. Hence $A \subseteq V_\alpha$, as required.

So it suffices to show that $D = \emptyset$. For reductio, suppose otherwise. By Foundation, there is some $B \in D \subseteq A$ such that $D \cap B = \emptyset$. If $x \in B$ then $x \in A$, since A is transitive, and since $x \notin D$, it follows that $\exists \delta\ x \subseteq V_\delta$. So now let

$$\beta = \subsetneq \{\, \delta \mid (\exists x \in b)(x \subseteq V_\delta \wedge (\forall \gamma < \delta) x \nsubseteq V_\gamma) \,\}.$$

As before, $B \subseteq V_\beta$, contradicting the claim that $B \in D$. □